Lloyd Roshan

ce24btech11036@iith.ac.in | lloydroshan07@gmail.com | +91 9778253960 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.26
PLACID VIDYA VIHAR SENIOR SECONDARY SCHOOL	Class XII (CBSE)	2024	97.60%
SANJOS METROPOLITAN SENIOR SECONDARY SCHOOL	Class X (CBSE)	2022	96.20%

SCHOLASTIC ACHIEVEMENTS

- Qualified the Indian Olympiad Qualifier in **Junior Science** Part I (**NSEJS**); shortlisted among **301 students** for IOQJS Part II and secured **State Top 1% in Kerala**, competing in an examination with **11,124 candidates** across the country.
- Qualified the Indian Olympiad Qualifier in **Mathematics** (**IOQM**) 2023-24 and advanced to the **Regional Mathematical Olympiad (RMO)**, the **second stage** of India's selection process for the **International Mathematical Olympiad** (**IMO**).
- Secured **100%** in **Physics** in the **AISSCE** (CBSE Class XII Board Examination) in the academic year 2023-24.
- Secured **100%** in **Science** in the **AISSE** (CBSE Class X Board Examination) in the academic year 2021-22.

SKILLS

- **Programming Languages:** C, C++, Python, MySQL.
- **Tools & Technologies:** Git, GitHub, VS Code, Overleaf, LaTeX
- **CS Fundamentals:** Object-Oriented Programming (OOPs), Data Structures & Algorithms, DBMS
- **ML Frameworks & Tools:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn.
- **Soft Skills:** Leadership, Teamwork, Communication, Problem Solving

PROJECTS

Housing Price Prediction -- Machine Learning Model

Python | *NumPy* | *Pandas* | *Matplotlib* | *Scikit-learn*

- Engineered a machine learning pipeline for housing price prediction, performing data preprocessing, exploratory data analysis (**EDA**), **feature engineering**, and **regression modeling** to identify key factors influencing property prices.
- Implemented and evaluated regression models using **Scikit-learn**, transforming raw housing data into meaningful features and training predictive models to generate accurate property price estimates.
- Analyzed **feature-target relationships** and model behavior through **statistical analysis** and **data visualization** using **Pandas** and **Matplotlib** to improve data quality and model performance.

Schedulix -- Constraint-Based Timetable Generation (Ongoing)

C++ | *Python* | *Backtracking* | *Graph Algorithms*

- Engineered a constraint-based school timetable generator using **backtracking** and **graph algorithms** to automate timetable construction while satisfying teacher, class, and scheduling constraints.
- Implementing **recursive constraint-based** search to generate feasible timetables, detect scheduling conflicts, and systematically explore valid assignments.
- Exploring **optimization techniques** to improve timetable quality by incorporating teacher availability, **consecutive-class constraints**, and **workload distribution** while minimizing scheduling conflicts.

Python-Based Beam Analysis Tool -- **Automated Structural Analysis**

Python | Structural Mechanics | Numerical Methods | Data Visualization

- Engineered a **Python-based beam analysis tool** to automate the generation of Shear Force Diagrams (**SFD**) and Bending Moment Diagrams (**BMD**)for user-defined loading and boundary conditions.
- Implemented computational procedures for structural analysis, handling different load and support configurations to determine the corresponding shear force and bending moment distributions along the beam.
- Integrated programming with core Civil Engineering concepts to **streamline repetitive structural calculations** and provide a **systematic framework** for analyzing beam response under varying loading conditions.

POSITIONS OF RESPONSIBILITY

Aeolus Racing, IIT Hyderabad - Core Member

- Contributed to the design and development of a Formula Student racing car for **Formula Bharat**, **representing IITH**; specialized in **Vehicle Dynamics** and suspension design, designing wishbones and rocker arms using **CAD** software.

Aero Club, IIT Hyderabad - Core Member

- Conducted an engaging session on fundamentals of **Aerodynamics and their Applications**, covering key concepts, practical aspects, followed by a Q&A session, with active participation from **200+** students at IIT Hyderabad.

RELEVANT COURSES

- **Computer Science:** Introduction to Programming, Data Structures & Applications
- **Mathematics & Statistics:** Calculus, Real Analysis, Linear Algebra, Differential Equations, Probability, and Statistics.